



Introduction to User Interface Design Module 1 ⁽⁶⁾

116U01E734
User Experience Design

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User Interface Design



What is user experience design?



What makes up user experience?



Evolution of UX Design,
Elements of user experience,
requirements, strategy



Design principles



User centric design



User Interface

- The layout or the structure of the elements on the screen as well as the functional behavior of such elements when manipulated by the user.
- The user interface design should be **user-friendly**



User Experience

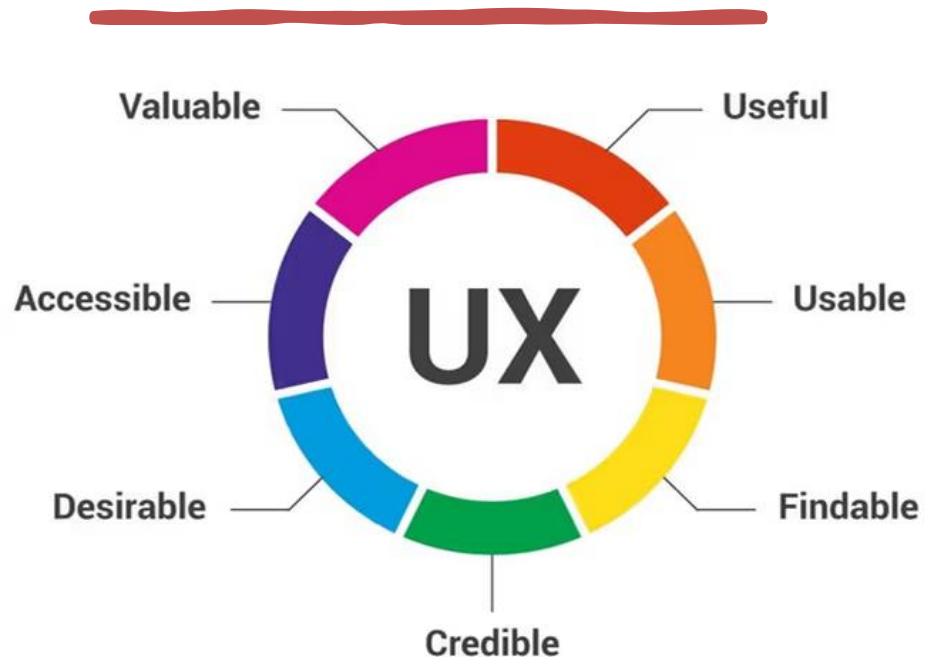
- User experience is **how the user feels while and *after* using the system**
- The user experience has to do with the ability of a user **to effectively and successfully manipulate** a software.
- Users rate them according to appearance, functionality, interactivity among others

Enhancing User Experience

- **Better user experience can be provided by**
 - Focusing on solving user problems
 - Meeting their needs in a way that is intuitive, efficient, and enjoyable
 - Following user-centered design which is consistence simple
 - Providing feedback
 - Supporting through Visual hierarchy
 - Thorough usability testing.

Enhancing User Experience

- A great user experience (UX) can be achieved for a product or service by making it:
 - Useful
 - Usable
 - Desirable
 - Findable
 - Accessible
 - Credible
 - Valuable



Enhancing User Experience

Useful

- The system should provide something ***needed*** by the user to solve some problems or complete task(s)
- The purpose could be
 - Entertainment
 - Communication
 - Education

Enhancing User Experience

Usable

- Enabling users to effectively and efficiently achieve their end objective with a product.
- Products can succeed if they are usable than those which are difficult to use
- Poor usability is often associated with the very first generation of a product its basically due to lack of understanding users' perspective and lack of through testing
- A computer game which requires 3 sets of control pads is unlikely to be usable as people, for the time being at least, only tend to have 2 hands.

Enhancing User Experience

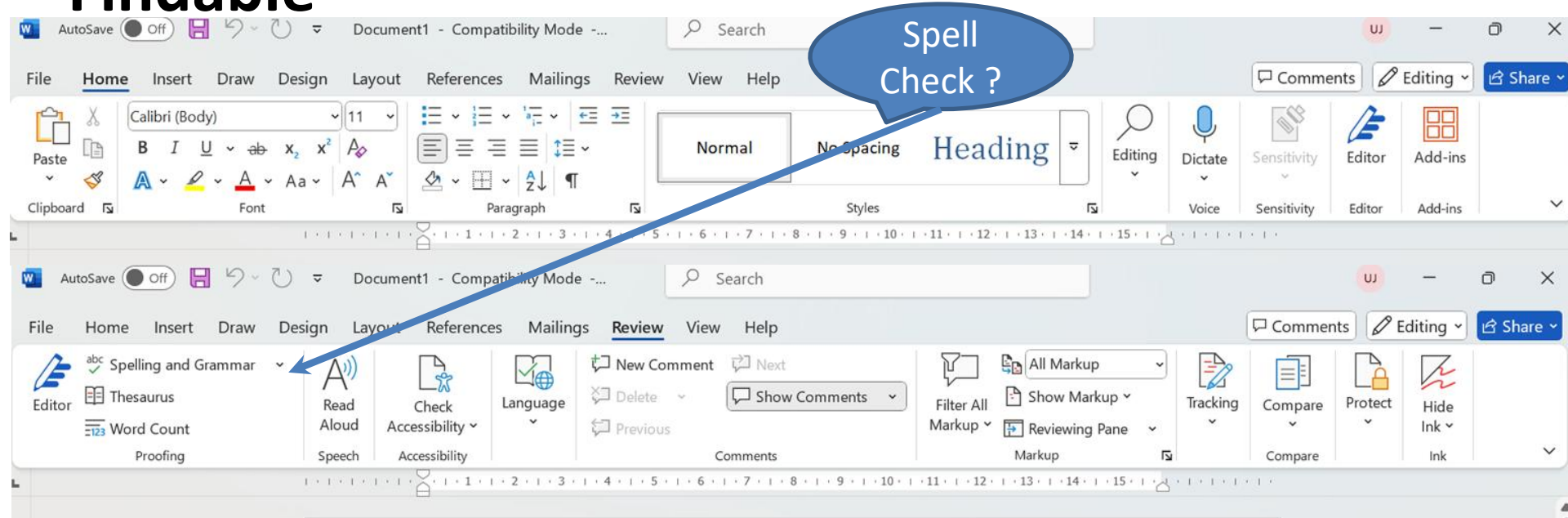
Desirable

- Conveyed in design through branding, image, identity, aesthetics and emotional design.
- The more desirable a product is – the more likely it is that the user who has it will brag about it and create desire in other users.



Enhancing User Experience

Findable



Findable refers to the idea that the product must be easy to find and in the instance of digital and information products; the content within them must be easy to find too.

If you cannot find a product, you're not going to buy it and that is true for all potential users of that product.

Findability is vital to the user experience of many products.

Enhancing User Experience

Accessible

- Accessibility is about providing an experience which can be accessed by users of a full range of abilities – this includes those who are disabled in some respect such as hearing loss, impaired vision, motion impaired or learning impaired.
- Design for accessibility will often find that the created products are easier for everyone to use not just those with disabilities.
- Accessible design is a legal obligation in many jurisdictions

Enhancing User Experience

Accessible

- Multilingual
- Physically/ Visually Challenged

- Having a customizable button function
- To make a visually challenged user, navigating a button function



Enhancing User Experience

Credible

- [Credibility](#) relates to the ability of the user to trust in the product that you've provided.
- Not just that it does the job that it is supposed to do but that it will last for a reasonable amount of time and that the information provided with it is accurate and fit-for-purpose.
- It is nearly impossible to deliver a user experience if the user thinks the product creator is a lying, clown with bad intentions – they'll take their business elsewhere instead.

Enhancing User Experience

Credible



Payment Options



Enhancing User Experience

Valuable

- the product must deliver value.
- Return of Investment
- Without value it is likely that any initial success of a product will eventually be undermined.
- The value is one of the key influences on purchasing decisions.
- A ₹100 product that solves a ₹ 10,000 problem is one that is likely to succeed; a ₹ 10,000 product that solves a ₹ 100 problem is much less likely to do so.

Comparison of User Interface (UI) and User Experience (UX)

UI	UX
Ui is what a user sees	UX is how a user feels
The UI designer is responsible for designing screens which users will interact with	UX designer makes sure the interaction is a pleasurable one
UI designer chooses Components and interaction styles used while performing operations	UX designer works on many factors including UI such as Latency rate, Overall time to perform operation etc.

a good UI contributes to a better UX

User Interface

Interaction styles :

- Command line
- Graphical User Interface
- Voice Activated
- Gesture based
- Brain Controlled
- Any other such as face recognition, biometric, iris
....

User Interface

Interaction styles :

– Command line

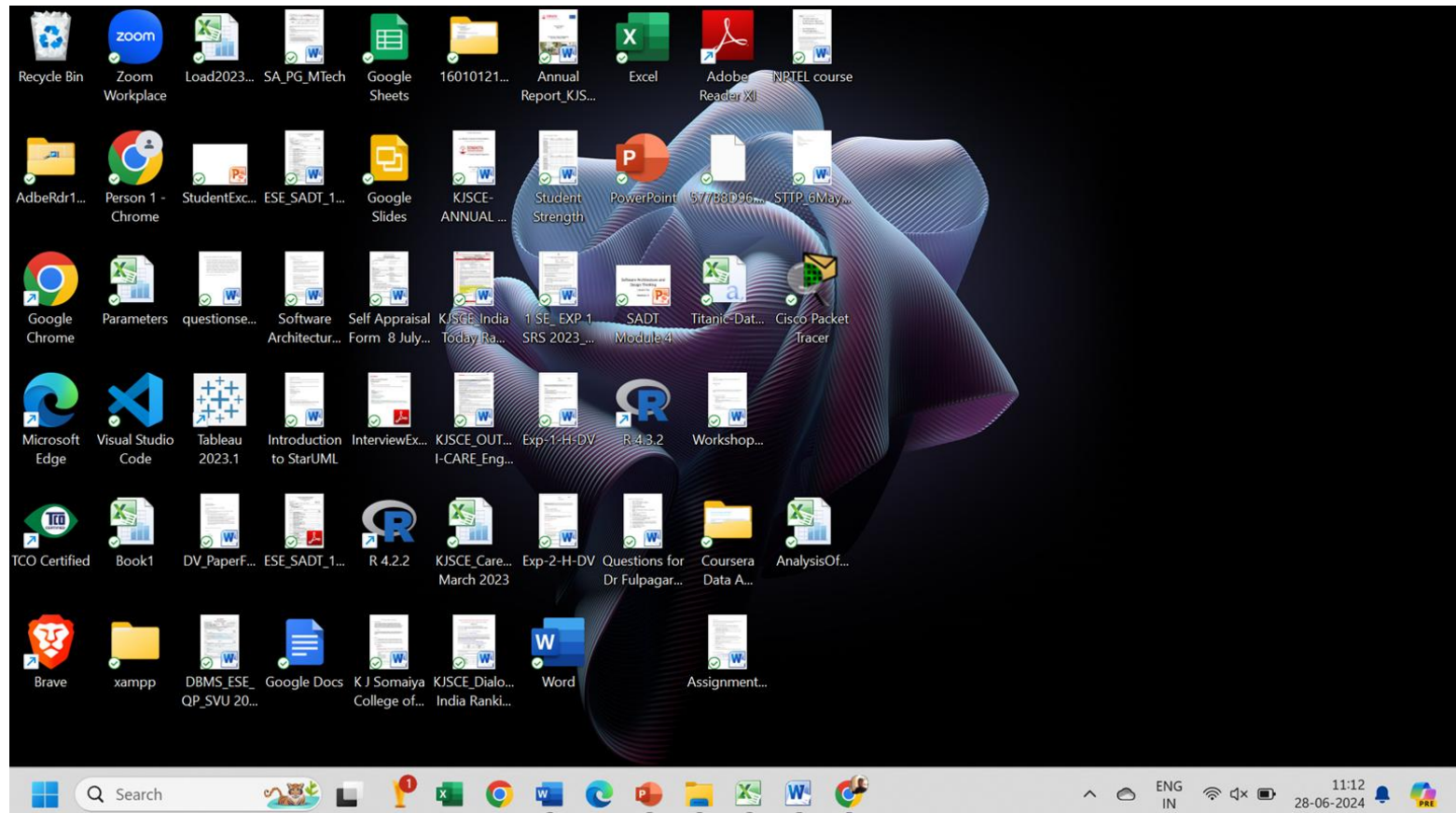


- Dry and boring
- Expects exact command
- Less user friendly
- Requires less resources

User Interface

Interaction styles :

– Graphical User Interface



User Interface

Interaction styles :

– Voice Activated

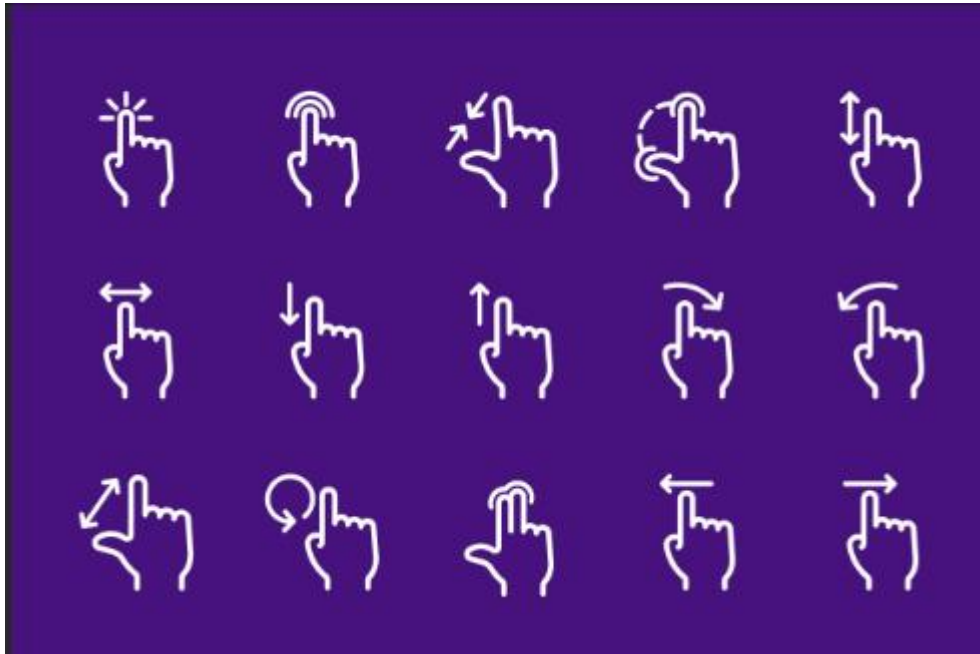


- Handsfree use
- Multitasking
- Continuous listening

User Interface

Interaction styles :

– Gesture based

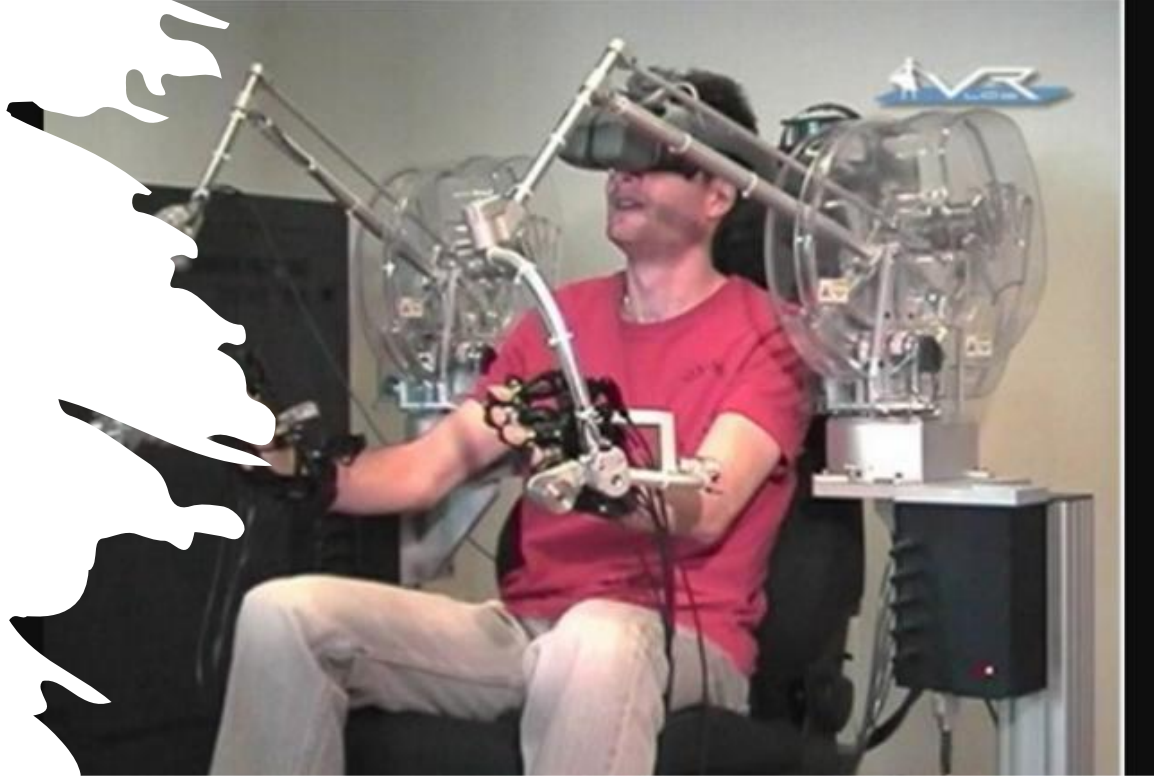


- Handsfree use
- Difficult to learn

User Interface

Interaction styles :

– Brain Controlled



- Physically Challenged
- Requires precision

User Interface

Interaction styles :

- Any other

- Face recognition:

- easier, no need to remember password
- Depend on light condition, more resources needed

- Biometric/ iris

- easier, no need to remember password
- Special scanner needed, issues with some human

User Interface

Interaction Components:

- Hardware
 - Software
-
- **Hardware**
 - Devices
 - Sensors
 - Actuators
 - **Software**
 - Screen components



User Interface

Interaction Components:

– Hardware

- Touch screen
- Mouse
- Joystick
- AR-VR





User Interface

Interaction Components:

– Software

- Screen components
 - Label
 - Textbox
 - Date picker
- Language of communication
-

User Experience

Definition:

- How a user feels while/after using the system
- Not restricted ONLY to computer system but ALL devices/ gadgets used these days



Basic Emotions



Joy



Trust



Fear



Surprise



Sadness



Disgust



Anger



Anticipation



User Experience

User Experience

- Name systems
 - You *prefer* for
 - ordering food online (Zomato and Swiggy, Uber Eats, Foodpanda, Domino's, and newer players like WAAYU.)
 - Ride hailing services (such as ola/ uber)

Justify

User Experience

Expectations from the system

- Intuitive
- Easy to use
- Easy to learn
- Allow to make error
- Look beautiful
- Reliable
- Efficient
- Deliver accurate results

And many more

User Experience

**Feelings / Emotions of users
while/ after using the system**

- Anxious
- Frustration
- Surprise
- Angry
- Helpless
- Happy
- Comfortable

**Overcoming these feelings
through**

- Good User Interface
- Intuitive
- Familiarity
- Use of latest
technology

User Experience

- Anxiety:
 - Results
 - Transfer of money
 - Booking of a cab
- Remedy:
 - Progress bar
 - Loading
 - Many more ...



Making up User Experience (UX)



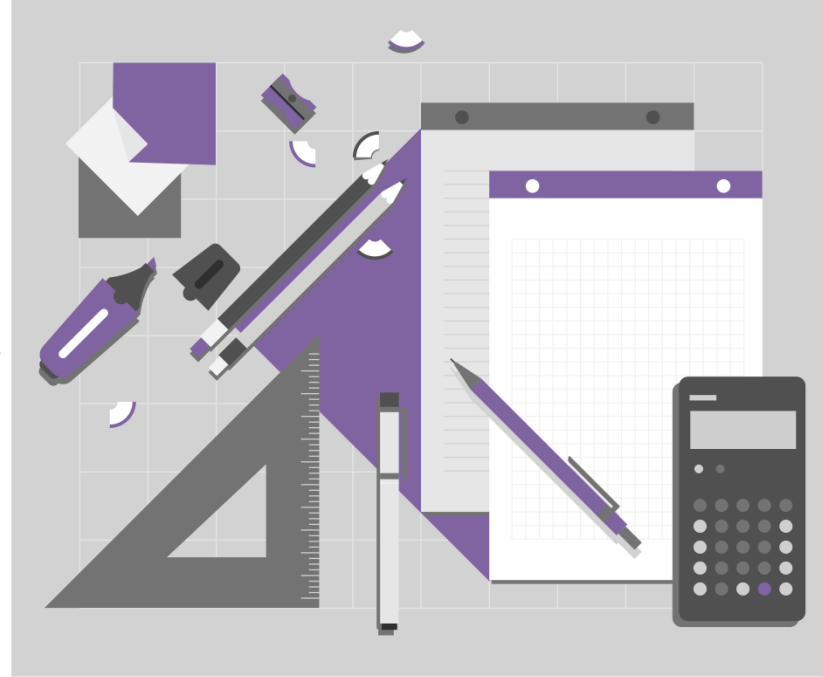
- Refers to the overall interaction that users have when using a product, service, website, or application.
- Considers every aspect of the user's interaction and is characterized by a seamless and enjoyable experience that meets the user's needs and expectations.
- Involves elements such as **intuitive navigation, aesthetically pleasing design, efficient functionality, and responsive customer support.**

Making up User Experience (UX)

- **Create user-centric solutions by:**
 - Understanding user behaviors
 - Conducting research
 - Implementing design principles to optimize the overall satisfaction and usability of the product

Making up User Experience (UX)

- **The Design Processes of User Experience**
 - **Researching**
 - **Analyze**
 - **Designing**
 - **Testing**
 - **Launching**



Making up User Experience (UX)

- **The Design Processes of User Experience**
 - **Researching**
 - Done thoroughly to attain accuracy from start to finish.
 - Gather fundamental information about their product or service — such as about the target users, their needs, and the data needed for their product. Subsequently, the collected data and information will become a basis for the next steps of the process.



Making up User Experience (UX)

- **The Design Processes of User Experience**
 - **Analyze**
 - analyze, plan, and map it out for further understanding.
 - Discover the possible ways and solutions to make interface
 - Visualize the feasible content and outlook of the UX that can meet the expectations of the users.



Making up User Experience (UX)



- **The Design Processes of User Experience**
- **Designing**
- Develop the whole product and create multiple versions to try and find the final best solution for the users.
- The chosen model must be accomplished correctly in all aspects, from layout to its function and making it accessible to the users.

Making up User Experience (UX)

- **The Design Processes of User Experience**
- **Testing**
- Testing the product first with real users to see how well it functions and meets the given criteria.
- Helps to identify any issues with the product and allows for improvements to be made immediately to ensure the final product work flawlessly before releasing it to the public.



Making up User Experience (UX)

- **The Design Processes of User Experience**
- **Launching**
- Launch and let the public know about it.
- The time for a company to see the public's verdict on whether it will succeed.
- The user's feedback is going to be a crucial factor if the product has satisfied them or if it needs to have further improvement.
- Essential to deliberately complete the process of making a UX and listen to the customers' feedback, if there's any.



Evolution of UX Design

- Evolution of UX Design

Elements of User experience

- Requirements, strategy
- Design principles,
- User centric design.

User Experience Requirements

- strategy
- Design principles,
- User centric design.

User Experience Strategy

- Design principles,
- User centric design.

Design Principles

UX design principles

1. Focus on the user
2. Consistency
3. Hierarchy
4. Context is key
5. Put the user in control
6. Accessibility
7. Usability testing

Design Principles

- **Focus on the user**
 - Study background of the end users
 - Gender, Age, Physical parameters- Height-weight, handedness, Educational background, cultural background, previous experience of using same/ similar systems
 - user-centric design

Design Principles

- **Consistency**
 - External Consistency
 - Appearance of the screens, components
 - Internal Consistency

Design Principles

- **Hierarchy**

Design Principles

- **Context is key**

Design Principles

- **Put the user in control**

Design Principles

- **Accessibility**

Design Principles

- **Usability testing**

Design Principles

- **Focus on the user**
 1. Emphasis
 2. Hierarchy
 3. Context is key
 4. Put the user in control
 5. Accessibility
 6. Usability testing

Design Principles

- **Emphasis**
 - **Fonts:** style, size
 - **Contrast:** color, size, shape, texture, or typography can help highlight specific elements.
 - **Empty space / White space.** Effective use of negative space can provide emphasis by isolating and highlighting specific elements.
 - **Motion.** Animation around areas to focus their attention is an unfailing strategy for ensuring emphasis
 - **Choose colors strategically:** to attract attention

Emphasis

Font



Contrast



White spaces



Motion



AM

Color Meanings	
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Red	Passion, confidence, energy, rage, danger
Purple	Royalty, wisdom, bravery, spirituality, uniqueness
Blue	Conservative, stable, calm, reliable, sad, aloof
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Green	Fresh, natural, fertile, abundant, peaceful
Yellow	Happy, optimistic, friendly, energy, caution
Orange	Creativity, confidence, excitement, happiness
Brown	Warm, earthy, reliable, stable, approachable
©LouiseM.com	
Black	Power, sophistication, mystery, evil, death
Gray	Neutrality, balance, calm, sadness, boredom
White	Purity, cleanliness, simplicity, neutrality
©LouiseM.com	
Pink	Romance, femininity, compassion, sweetness

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User Centric Design